Intelligent Agent

Dr. Nirmala Shinde Baloorkar

Assistant Professor

Department of Computer Engineering. KJSSE

SOMAIYA VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering

Somaiya TRUST

Last time's key concepts

- Agents : An agent is anything that can be viewed as –
  - perceiving its environment through sensors and
  - acting upon that environment through actuators.
- "Rational Agent" as the organizing theme
  - Acts to maximize expected performance measure
- Task Environment : PEAS

Outline

- Properties of Task environments

What information do the sensors provide?

- Fully observable/Partially observable :
  - A task environment is (effectively) fully observable iff the sensors detect the complete state of the environment
  - ”relevant” depends on the performance measure
  - no need to maintain internal state to keep track of the environment
- Partially observable :
  - An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data .
- Unobservable:
  - If the agent has no sensors at all.

Does the performance depend on other agents?

- Single agent:
  - . An agent operates independently in an environment, and its performance is unaffected by the presence or actions of other agents.
- Competitive multi-agent:
  - Improving the performance of one agent directly reduces the performance of another.
- Cooperative multi-agent:
  - Improving the performance of one agent also improves the performance of the others.

What effect do the actions have?

- Deterministic:
  - Outcome of actions (next state of the environment) is completely(fully) determined by the current state.
- Non-deterministic:
  - Each action has a set of possible outcomes.
- Stochastic:
  - There is a probability distribution over possible outcomes

How do the percept change over time?

- Episodic:
  - Agent receives a **single percept** per episode.
  - It's action in this episode doesn't affect future percepts.
  - E.g. classification
- Sequential:
  - Agent receives a **sequence of percepts**.
  - The current action affects future percepts.
  - E.g. navigation, game playing

Is the world discrete/continuous?

- Continuous:
  - Time, percepts and actions are continuous.
  - E.g. driving car
- Discrete:
  - Time, percepts and actions are discrete.
  - E.g. playing a board game

How does the environment change over time?

- Static:
  - Environment doesn't change unless agent performs an action
- Dynamic:
  - Environment changes even when the agent doesn’t do anything
- Semi-dynamic:
  - Environment is static, but agent’s performance score changes over time

What is the agent's knowledge about the environment?

- Known:
  - Agent knows **all the rules that** hold in the environment. Can predict outcomes from complete observations.
- Unknown:
  - Agent **doesn't know the rules** of the environment. Can't predict outcomes

Summary

- Properties of task environment are

1. Observable
2. Agent
3. Deterministic
4. Episodic
5. Static
6. Discrete
7. Known

PEAS for Poker

- **Performance Measure**:
  - Winning the game (accumulating the most chips).
  - Making profitable decisions (maximizing expected value of bets).
- **Environment**:
  - The poker table.
  - Other players.
  - The deck of cards.
- **Actuators**:
  - Actions such as betting, folding, raising, and calling.
- **Sensors**:
  - Visual perception of own cards and community cards.
  - Observing opponents' actions and behaviors.

Example – **Poker**

- Observable:
- Agents:
- Deterministic
- Episodic:
- Static
- Discrete:

- Partially
- Multi
- Stochastic
- Sequential
- Static
- Discrete

Example – **CrossWord Puzzle**

- Observable:
- Agents:
- Deterministic
- Episodic:
- Static
- Discrete:

- Fully
- Single
- Deterministic
- Sequential
- Static
- Discrete

Example –
Taxi driving

- Observable:
- Agents:
- Deterministic
- Episodic:
- Static
- Discrete:

- Partially
- Multi
- Stochastic
- Sequential
- Dynamic
- Continuous

Example –
Image Analysis

- Observable:
- Agents:
- Deterministic
- Episodic:
- Static
- Discrete:

- Fully
- Single
- Deterministic
- Episodic
- Semi
- Continuous

Example –
Recipe Execution

• Observable:	• Fully
• Agents:	• Single
• Deterministic	• Deterministic
• Episodic:	• Sequential
• Static	• Static
• Discrete:	• Continuous

Quiz

In which type of environment does the agent know all the rules and can predict outcomes from complete observations?

A. Known

B. Unknown

C. Partially Observable

D. Unobservable

Quiz

Which of the following best describes a **partially observable** environment?

A. The agent has no sensors at all

B. The agent can see the entire state of the environment

C. The agent has noisy or incomplete sensor data

D. The agent's actions do not affect future percepts

Quiz

Which property of the task environment is characterized by the environment changing even when the agent doesn't do anything?

A. Static

B. Dynamic

C. Semi-dynamic

D. Deterministic

Quiz

Which of the following best describes a **stochastic** environment?

A. The outcome of actions is fully determined by the current state

B. Each action has a set of possible outcomes

C. There is a probability distribution over possible outcomes

D. The environment does not change unless the agent performs an action

Exercise

- Develop a PEAS description of the task environment for a face-recognition agent
- Describe the task environment
- Select a suitable agent design for the face recognition agent

Question ?